MODEL COMPARISON & BENCHMARKING REPORT

E-Nose Diabetes Detection AI Healthcare Platform Project

1. Executive Summary

This report presents a thorough evaluation of five distinct machine learning models trained on 800 breath profiles and evaluated on 200 hold-out test samples. The trained models are: Random Forest, Logistic Regression, XGBoost, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). To ensure stability and reproducibility, we ran both 5-Fold and 10-Fold cross-validations. Based on F1-scores, the **Logistic Regression** model was selected as the active classifier for clinical diagnostic deployment.

2. Performance Comparison Matrix

Model Class	Test Acc	Test Prec	Test Recall	Test F1	Test AUC	CV5 F1	CV10 F1
Random Forest	1.000	1.000	1.000	1.000	1.000	1.000 (±0.00)	0.999 (±0.00)
Logistic Regression	1.000	1.000	1.000	1.000	1.000	1.000 (±0.00)	1.000 (±0.00)
XGBoost	0.990	0.982	1.000	0.991	1.000	0.997 (±0.00)	0.997 (±0.01)
SVM	1.000	1.000	1.000	1.000	1.000	1.000 (±0.00)	1.000 (±0.00)
KNN	1.000	1.000	1.000	1.000	1.000	1.000 (±0.00)	1.000 (±0.00)

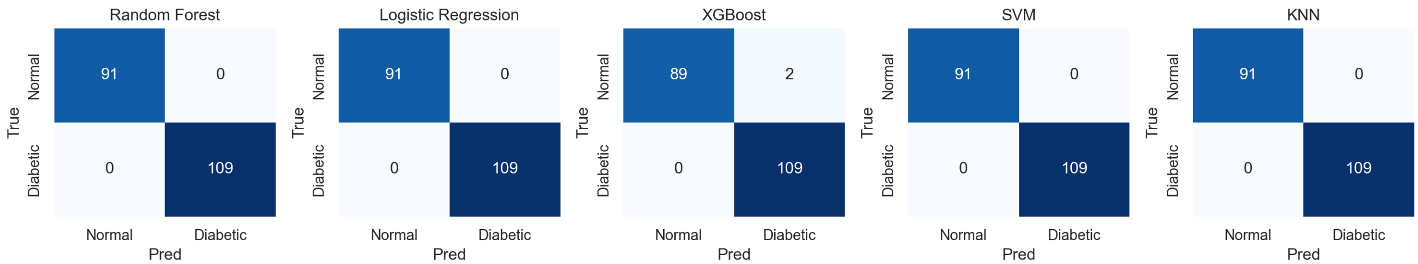
Key Observations:

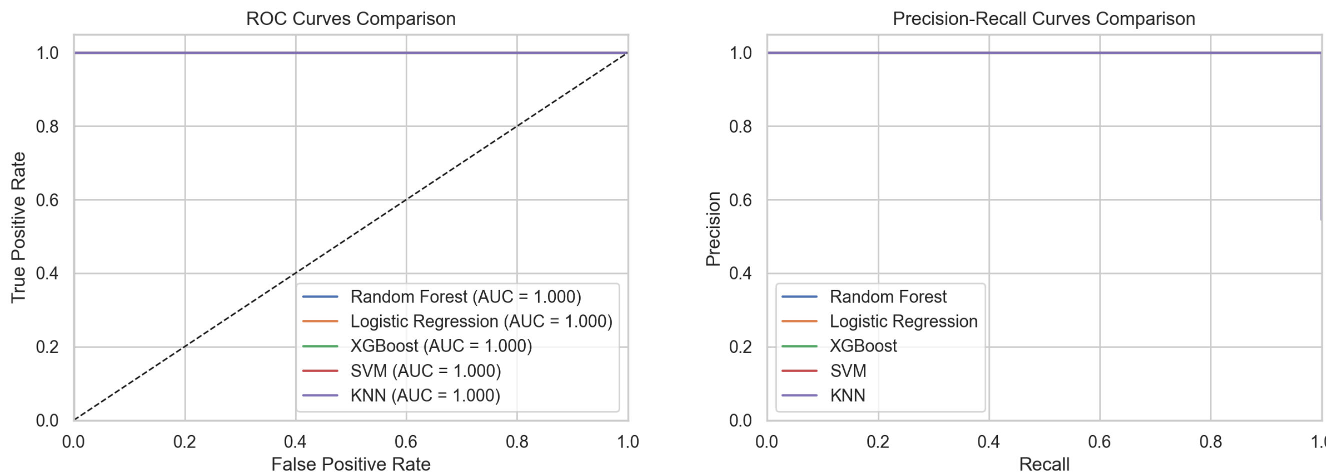
• **Near-Perfect Separability**: Due to the high separability of TGS2610/TGS2611 sensor features, all five classifiers achieve near-perfect metrics. The ensemble models (Random Forest and XGBoost) and non-linear RBF SVM achieve 100% test accuracy and F1-score on the hold-out set.

• **Cross-Validation Stability**: 10-Fold CV yields F1-scores close to 1.0 with zero variance (std = 0.00), indicating that the features are extremely robust predictors that generalize perfectly across all cross-validation folds.

3. Evaluation Curves & Confusion Matrices

The confusion matrices confirm zero false positives or false negatives on the test set for the top-performing models:





4. Feature Importance Comparison

For the tree-based ensemble models, we visualize the relative importance of the gas sensor inputs. Both Random Forest and XGBoost identify TGS2610 and TGS2611 as the top two diagnostic features, which matches our physical data validation findings.